Lab 2 RGB LED

Introduction

RGB LED modules can emit various colors of light. Three LEDs of red, green, and blue are packaged into a transparent or semitransparent plastic shell with four pins led out. The three primary colors, red, green, and blue, can be mixed and compose all kinds of colors by brightness, so you can make an RGB LED emit colorful light by controlling the circuit.

Principle

In this experiment, we will use PWM technology to the brightness of RGB.

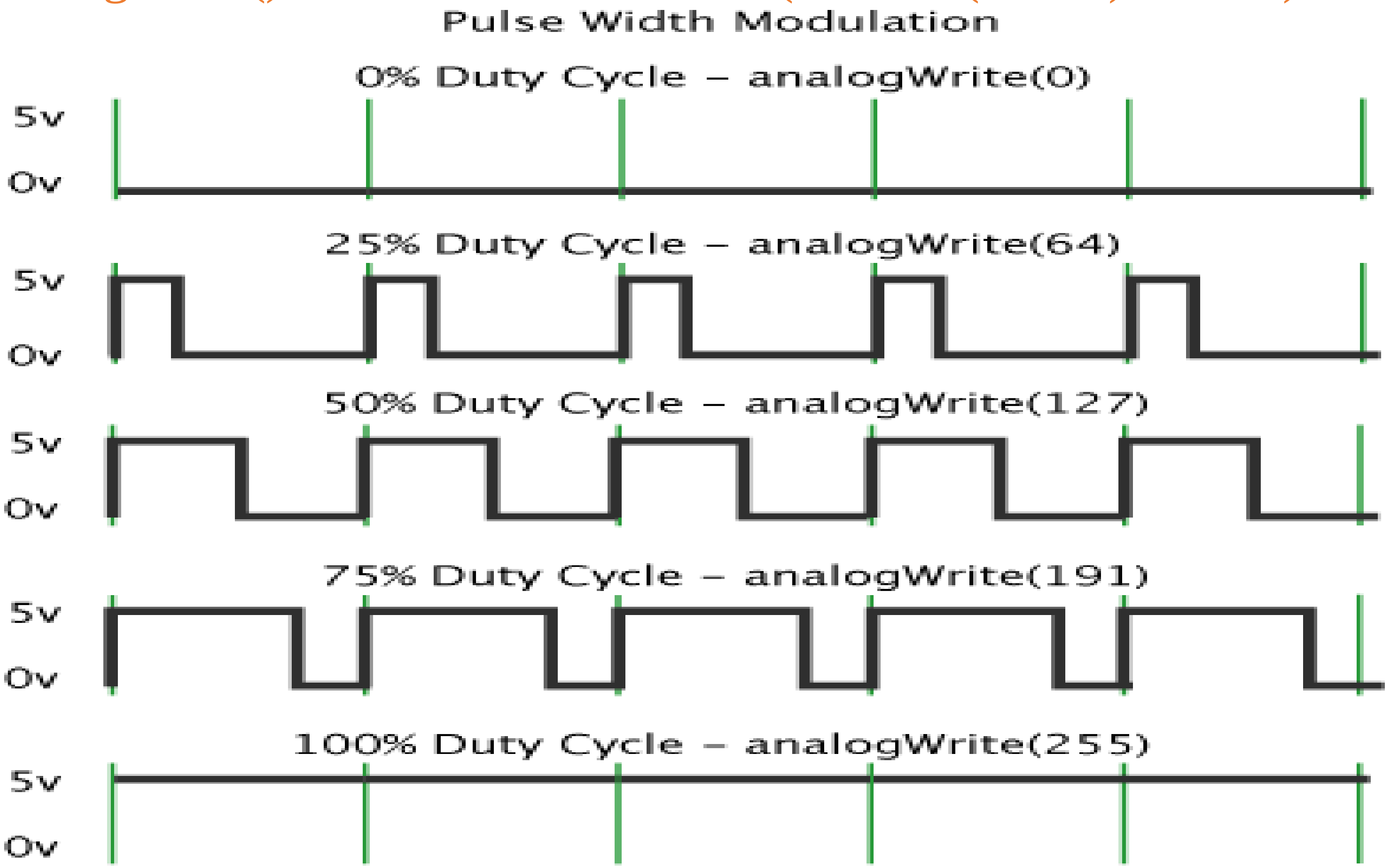
PWM (Pulse Width Modulation)

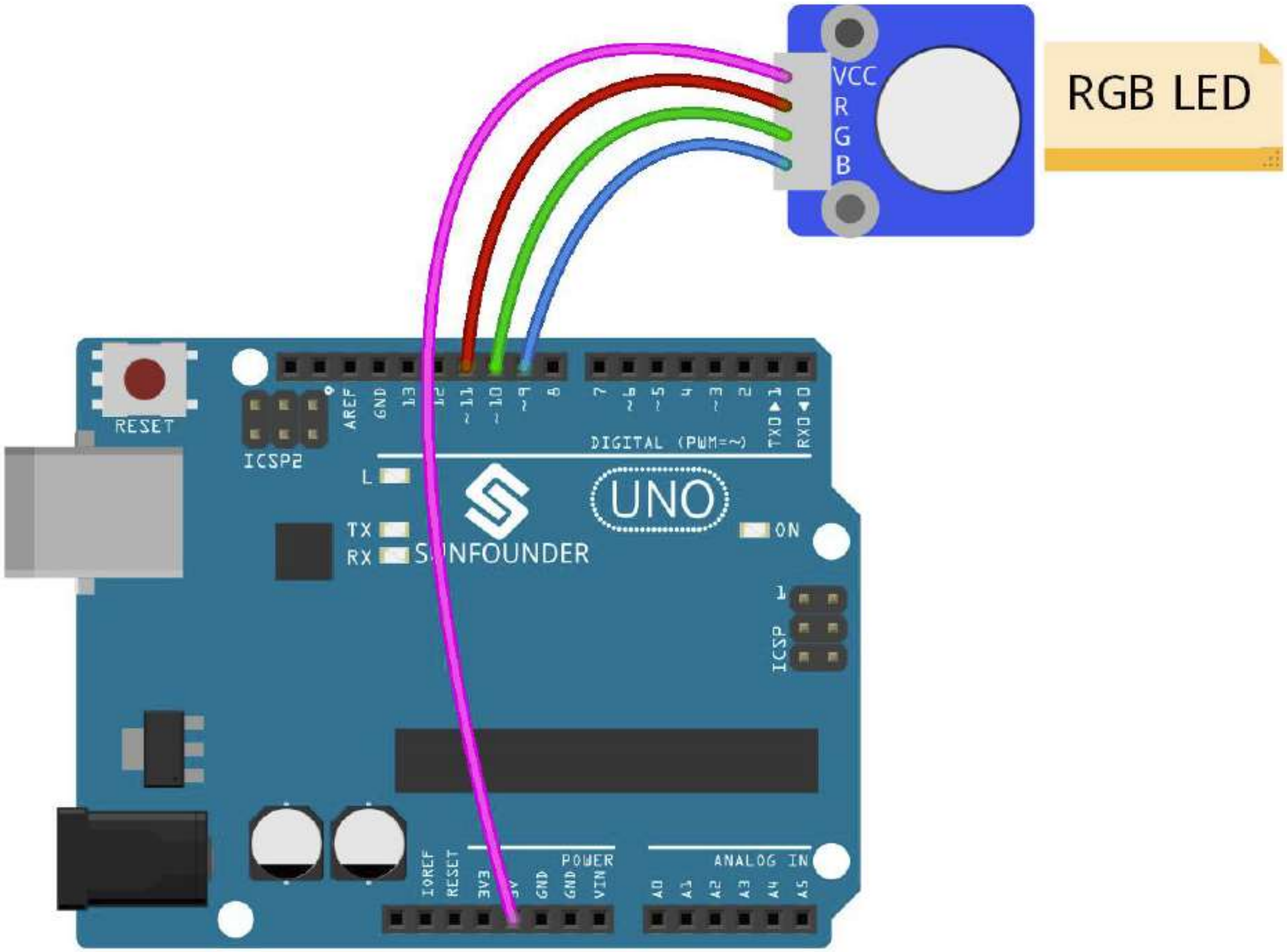
- PWM is a technique for getting **analog** results with **digital** signal.
- Digital control is used to create a square wave, a signal switched between on and off.
- This on-off pattern can simulate voltages in between full on (5V) and off (0V) by changing the portion of the time the signal spends on versus the time that the signal spends off.
- The duration of "on time" is called the **pulse width**.
- To get varying analog values, change or modulate the pulse width.
- When repeating this on-off pattern fast enough with an LED for example, the result is as if the signal is a steady voltage between 0 and 5v controlling the brightness of the LED.

PWM (Pulse Width Modulation)

-Arduino’s PWM frequency is at about 500Hz(T = 2ms).

-analogWrite() is on a scale of 0 -255 (Vout = (Ton/T)*Vmax)





RGB LED
VCC
R
G
B
RESET
ICSP2
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX▶1
RX◀0
DIGITAL (PWM=~)
L
TX
RX
SUNFOUNDER
UNO
ON
ICSP
1
IOREF
RESET
3V3
5V
GND
GND
VIN
POWER
ANALOG IN
A0
A1
A2
A3
A4
A5

Reference: RGB Color Table

Color	HTML / CSS Name	Hex Code#RRGGBB	Decimal Code(R,G,B)
	Black	#000000	(0,0,0)
	White	#FFFFFF	(255,255,255)
	Red	#FF0000	(255,0,0)
	Lime	#00FF00	(0,255,0)
	Blue	#0000FF	(0,0,255)
	Yellow	#FFFF00	(255,255,0)
	Cyan / Aqua	#00FFFF	(0,255,255)
	Magenta / Fuchsia	#FF00FF	(255,0,255)
	Silver	#C0C0C0	(192,192,192)
	Gray	#808080	(128,128,128)
	Maroon	#800000	(128,0,0)
	Olive	#808000	(128,128,0)
	Green	#008000	(0,128,0)
	Purple	#800080	(128,0,128)
	Teal	#008080	(0,128,128)
	Navy	#000080	(0,0,128)

```
/************************************************************************/
 * Function: see RGB LED flash red, green and blue first, and then change to red,
orange, yellow, green, blue, indigo and purple.
*************************************************************************/
const int redPin = 11;  // R petal on RGB LED module connected to digital pin 11
const int greenPin = 10;  // G petal on RGB LED module connected to digital pin 10
const int bluePin = 9;  // B petal on RGB LED module connected to digital pin 9
/**************************************************************************/
void setup()
{
  pinMode(redPin, OUTPUT); // sets the redPin to be an output
  pinMode(greenPin, OUTPUT); // sets the greenPin to be an output
  pinMode(bluePin, OUTPUT); // sets the bluePin to be an output
}
```

```
//code below is for common-anode RGB LED
void loop()  // run over and over again
{  // Example basic colors:
  color(0,255,255); // turn the RGB LED red
  delay(1000); // delay for 1 second
 color(255,255,0); // turn the RGB LED blue
  delay(1000); // delay for 1 second
  // Example blended colors:
color(0,128,255); // turn the RGB LED orange
  delay(1000); // delay for 1 second
  color(0,0,255); // turn the RGB LED yellow
  delay(1000); // delay for 1 second
  color(255,0,0); // turn the RGB LED  indigo
  delay(1000); // delay for 1 second
  color(127,255,127); // turn the RGB LED purple
  delay(1000); // delay for 1 second
}
```

```
 void color (unsigned char red, unsigned char green, unsigned char blue)
// the color generating function
{
  analogWrite(redPin, red);
  analogWrite(bluePin, blue);
  analogWrite(greenPin, green);
}
```